Signals and Systems
Assignment-5: Solution

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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Exercise 1

- The Fourier transform of $x(t)$ is given by:

$$X(f) = \mathrm{rect}\left(\frac{f}{50}\right)$$

- $X(f)$ is bandlimited to the interval $[-25 \;\; 25]$ Hz resulting in $f_{max} = 25$ Hz (maximum frequency in the spectrum).
- Consequently:

$$f_s = 10 f_{\mathrm{Nyq}} = 10(2 f_{max}) = 20 f_{max} = 500 \;\; \mathrm{Hz}$$

Exercise 2

- The signal $x(t)$ has more rapid variations. Consequently, it occupies a larger bandwidth and, hence, needs to be sampled more often.

Exercise 3 (1)

- The Fourier transform of $x(t)$ is given by:

$$X(f) = \frac{1}{2}\delta(f - 100) + \frac{1}{2}\delta(f + 100)$$

- The sampling frequency is given by:

$$f_s = \frac{3}{4} f_{\mathrm{Nyq}} = \frac{3}{4}(2 \times 100) = 150 \ \ (\mathrm{Hz})$$

  Note that $f_s < f_{\mathrm{Nyq}}$ and aliasing will occur in this case.

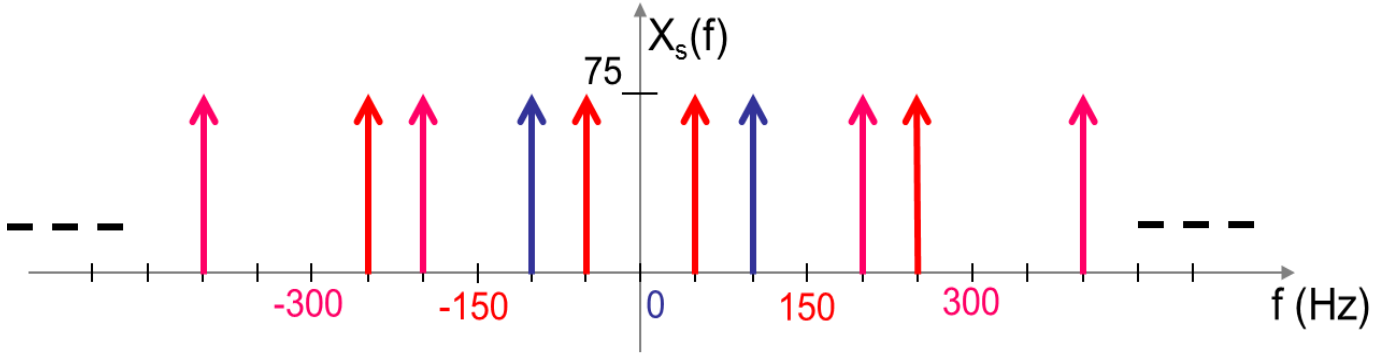
- The FT of the sampled signal is given by:

$$X_s(f) = \frac{1}{T_s} \sum_n X(f - nf_s) = 150 \sum_n X(f - nf_s)$$

  where $T_s = \frac{1}{f_s} = \frac{1}{150}$ (s).

Exercise 3 (2)

- $X_s(f)$ is shown in the figure below (1 unit on the x-axis corresponds to 50 Hz):



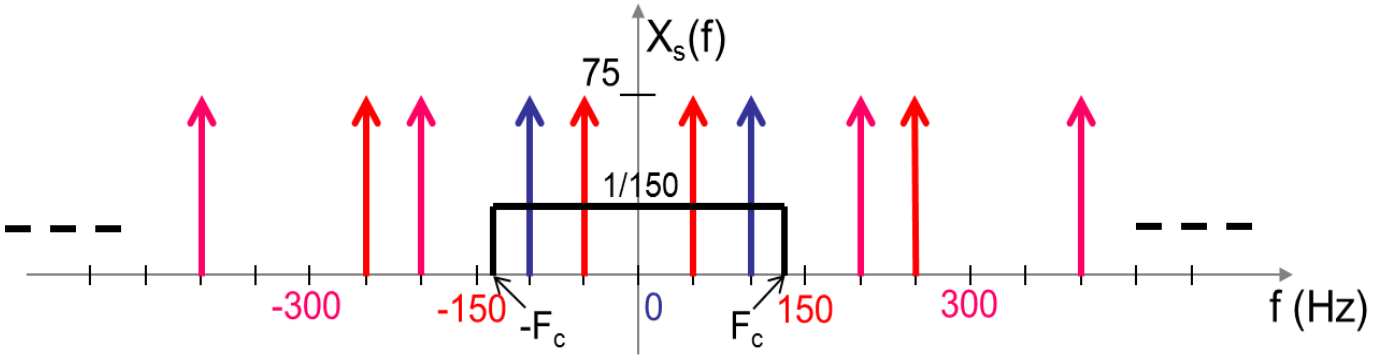
- The frequency components of the n-th replica are at:

$$nf_s \pm 100 = 150n \pm 100 \ \ (\mathrm{Hz})$$

  - For $n = 0$ (in blue): ± 100 Hz.
  - For $n = 1$ (in red): $150 \pm 100 = \{50, 250\}$ Hz.
  - For $n = -1$ (in red): $-150 \pm 100 = \{-50, -250\}$ Hz.
  - For $n = 2$ (in magenta): $300 \pm 100 = \{200, 400\}$ Hz.
  - For $n = 2$ (in magenta): $-300 \pm 100 = \{-200, -400\}$ Hz.

Exercise 3 (3)

- The LPF has a cutoff frequency between $f_{max} = 100$ Hz and $f_s = 150$ Hz. The gain of the filter is $\frac{1}{f_s} = \frac{1}{150}$.



- The frequency components that will pass through the filter are at ± 50 Hz and ± 100 Hz:

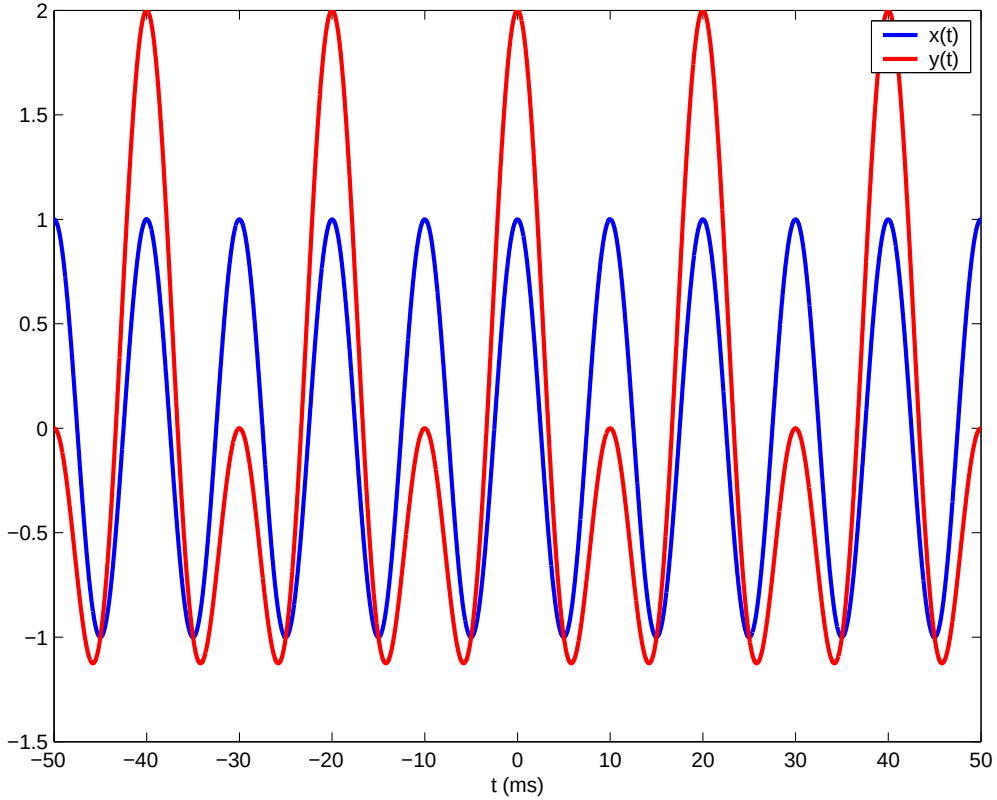
$$Y(f) = \frac{75}{150}\left[\delta(f-50) + \delta(f+50) + \delta(f-100) + \delta(f+100)\right]$$

$$\Rightarrow \quad y(t) = \cos(100\pi t) + \cos(200\pi t)$$

- Note that $y(t) \neq x(t)$ because of aliasing.

Exercise 3 (4)

- The signals $x(t)$ and $y(t)$ are shown in the figure below:



Exercise 4 (a)

- $x(t)$ contains the two frequencies of 100 Hz and 300 Hz.
- Consequently, the Nyquist rate is given by:

$$f_N = 2f_{max} = 2 \times 300 = 600 \quad \text{(Hz)}$$

- Implying that the sampling frequency is given by:

$$f_s = 3f_N = 3 \times 600 = 1800 \quad \text{(Hz)}$$

- The n-th sample is given by:

$$\begin{aligned} x[n] = x(nT_s) &= x\left(\frac{n}{1800}\right) \\ &= \cos\left(200\pi\frac{n}{1800}\right) + 3\cos\left(600\pi\frac{n}{1800}\right) \\ &= \cos\left(\frac{n\pi}{9}\right) + 3\cos\left(\frac{n\pi}{3}\right) \end{aligned}$$

Exercise 4 (b.1)

- The first four samples are given by:

$$x[0] = \cos(0) + 3\cos(0) = 4$$
$$x[1] = \cos\left(\frac{\pi}{9}\right) + 3\cos\left(\frac{\pi}{3}\right) = 2.4397$$
$$x[2] = \cos\left(\frac{2\pi}{9}\right) + 3\cos\left(\frac{2\pi}{3}\right) = -0.734$$
$$x[3] = \cos\left(\frac{3\pi}{9}\right) + 3\cos\left(\frac{3\pi}{3}\right) = -2.5$$

- On the other hand, the maximum and minimum values of $x(t)$ are:

$$\max[x(t)] = 4$$
$$\min[x(t)] = -4$$

implying that the dynamic range of the signal is:

$$D = \max[x(t)] - \min[x(t)] = 8$$

Exercise 4 (b.2)

- For 3-bit ADC, the number of quantization levels is: $q = 2^3 = 8$.
- The width of each quantization interval is:

$$\Delta = \frac{D}{q} = \frac{8}{8} = 1$$

  - In this case, the quantization intervals are given by: $[-4 \quad -3]$, $[-3 \quad -2]$, $[-2 \quad -1]$, $[-1 \;\; 0]$, $[0 \;\; 1]$, $[1 \;\; 2]$, $[2 \;\; 3]$ and $[3 \;\; 4]$.
  - Note that negative samples will belong to the first four intervals while positive samples will belong to the last four intervals.

Exercise 4 (b.3)

- The codewords associated with the different intervals are given in the table below:

Interval	codeword
[3 4]	111
[2 3]	110
[1 2]	101
[0 1]	100
[-1 0]	011
[-2 -1]	010
[-3 -2]	001
[-4 -3]	000

- The codewords associated with the first four samples are:

$$
\begin{aligned}
x_0 = 4 \quad &\Rightarrow \quad c_0 = 111 \\
x_1 = 2.43 \quad &\Rightarrow \quad c_1 = 110 \\
x_2 = -0.73 \quad &\Rightarrow \quad c_2 = 011 \\
x_3 = -2.5 \quad &\Rightarrow \quad c_3 = 001
\end{aligned}
$$

Exercise 4 (b.4)

- For 4-bit ADC, the number of quantization levels is: $q = 2^4 = 16$.
- The width of each quantization interval is:

$$\Delta = \frac{D}{q} = \frac{8}{16} = 0.5$$

- The codewords associated with the different intervals are given in the table below:

Interval	codeword
[3.5 4]	1111
[3 3.5]	1110
[2.5 3]	1101
[2 2.5]	1100
[1.5 2]	1011
[1 1.5]	1010
[0.5 1]	1001
[0 0.5]	1000
[-0.5 0]	0111
[-1 -0.5]	0110
[-1.5 -1]	0101
[-2 -1.5]	0100
[-2.5 -2]	0011
[-3 -2.5]	0010
[-3.5 -3]	0001
[-4 -3.5]	0000

Exercise 4 (b.5)

- The codewords associated with the first four samples are:

$$
\begin{aligned}
x_0 = 4 \quad &\Rightarrow \quad c_0 = 1111 \\
x_1 = 2.43 \quad &\Rightarrow \quad c_1 = 1100 \\
x_2 = -0.73 \quad &\Rightarrow \quad c_2 = 0110 \\
x_3 = -2.5 \quad &\Rightarrow \quad c_3 = 0010 \text{ or } 0011
\end{aligned}
$$

  - Note that the last result follows since x_3 belongs to two intervals: $[-3 \ -2.5]$ and $[-2.5 \ -2]$.

Exercise 4 (c)

- For 3-bit ADC, the mean square error is given by:

$$E = \frac{\Delta^2}{12} = \frac{1}{12} \approx 8.33\%$$

- For 4-bit ADC, the mean square error is given by:

$$E = \frac{\Delta^2}{12} = \frac{0.5^2}{12} = \frac{1}{48} \approx 2.08\%$$

- Note that adding one quantization bit reduced the mean square error by a factor of 4.

Exercise 4 (d)

- For 3-bit ADC, the size of the buffer is given by:

$$\begin{aligned} S &= f_s \times n \times T \\ &= (1800 \ \text{ samples/s}) \times (3 \ \text{ bits/sample}) \times (60 \ \text{ s}) = 324000 \ \text{ bits} \end{aligned}$$

- For 4-bit ADC, the size of the buffer is given by:

$$\begin{aligned} S &= f_s \times n \times T \\ &= (1800 \ \text{ samples/s}) \times (4 \ \text{ bits/sample}) \times (60 \ \text{ s}) = 432000 \ \text{ bits} \end{aligned}$$

- Note that, compared to the 3-bit ADC, the 4-bit ADC results in a MSE that is 4 times smaller at the expense of a storage space that is $4/3$ times larger.